Introduction to Biomaterials

Lecture [6]

Ceramic Biomaterials: Structure, Properties, and Biomedical Applications

Goals

- After reading this chapter, students will understand the following.
- The general definition of a ceramic.
- Common properties of ceramics. Different classifications used for ceramics.
- Properties of different bioceramics.
- Different technologies used for fabricating nanoceramics.

Ceramics as Biomaterials

- **Ceramics** have been used in **medicine for centuries**.
- **Early** examples include **artificial teeth** found in **ancient Egyptian mummies**.
- Today, ceramics are used in many biomedical applications because they can be:
  - Hard
  - Wear resistant
  - Chemically stable
  - Biocompatible
  - Similar to bone minerals in some cases



What is a Ceramic?

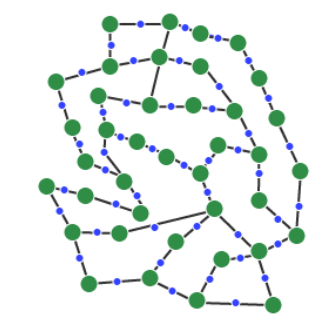
- A ceramic is an:
  - **Inorganic, non-metallic material**
- It usually contains :
  - **Metallic elements**
  - **Non-metallic elements**
- Ceramics are different from metals and polymers because their atoms are mainly bonded by:
  - **Ionic bonds**
  - **Covalent bonds**

Ceramics vs Metals vs Polymers

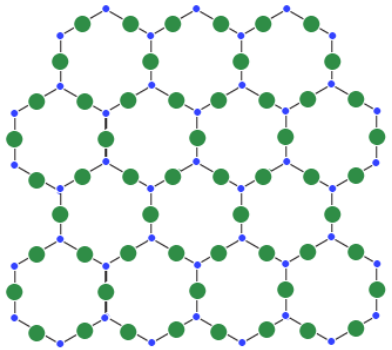
Material type	Main bonding	General behavior
Metals	Metallic bonding	Strong, ductile, conductive
Polymers	Covalent bonds in chains	Flexible, lightweight
Ceramics	Ionic + covalent bonding	Hard, brittle, stable

Atomic Structure of Ceramics

- Ceramics can have **two main** atomic arrangements:
- **Amorphous structure**
  - Random atomic arrangement
  - No long-range order
  - Example: **glass**
- **Crystalline structure**
  - Atoms arranged in an ordered pattern
  - Has long-range order
  - Example: **porcelain**

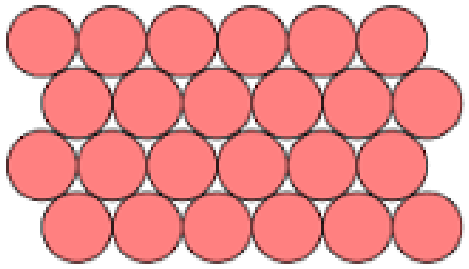


Amorphous Solids

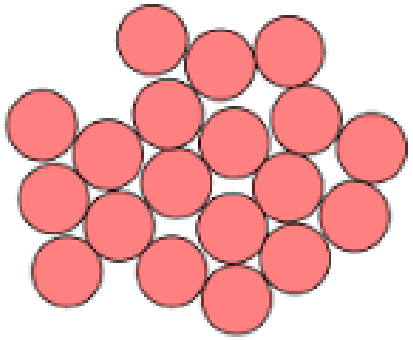


Crystalline Solids

Amorphous vs Crystalline Ceramics



Crystalline



Amorphous

Type	Atomic arrangement	Example
Amorphous ceramic	Random / short-range order only	Glass
Crystalline ceramic	Regular / long-range order	Porcelain

General Properties of Ceramics

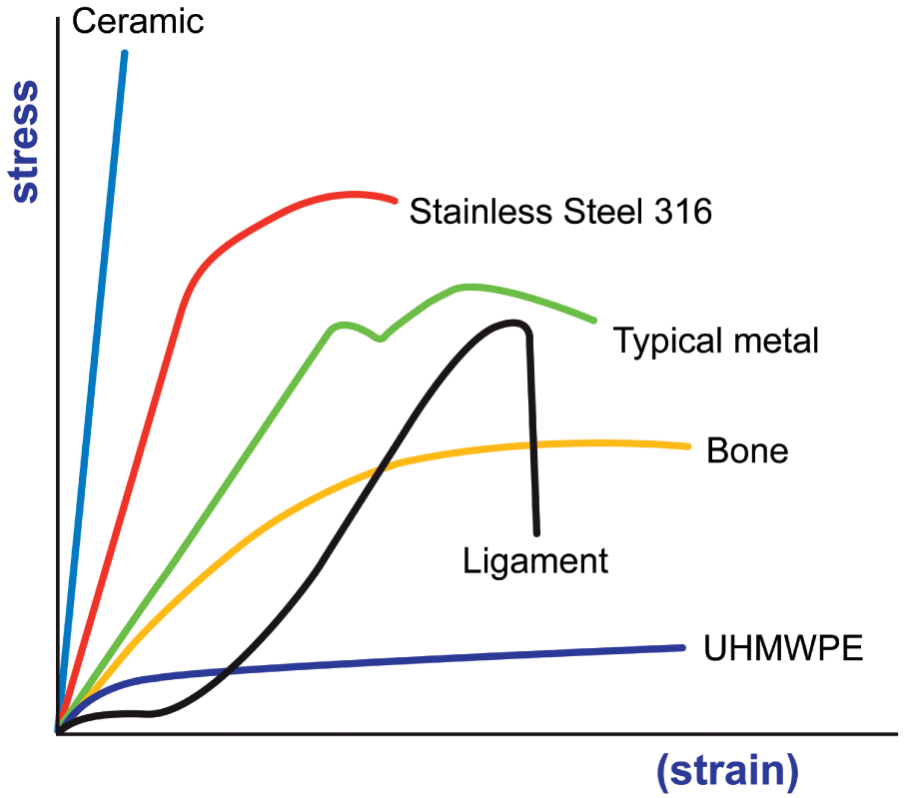
- Ceramics have **strong atomic bonds**.
- These bonds are mainly:
  - **Ionic bonds**
  - **Covalent bonds**
- Because of these strong bonds, ceramics are usually:
  - Hard
  - Chemically stable
  - Heat resistant
  - Resistant to wear and abrasion

Why Ceramics Are Hard and Stable

- **Strong ionic** and **covalent bonds** give ceramics **high stability**.
- This leads to important properties:
  - High hardness
  - High elastic modulus
  - Good dimensional stability
  - High resistance to chemical attack
  - High resistance to heat

Mechanical Behavior of Ceramics

- Ceramics are:
  - Strong in compression
  - Weak in tension
  - Brittle
- This means ceramics can c**arry high compressive loads**, **but** they may **fail when pulled**, **bent**, or **impacted**.
- → Ceramics can **resist squeezing**, but they do **not tolerate stretching** very well.



Brittleness of Ceramics

- Ceramics **do not** usually **undergo plastic deformation** like metals.
- Instead, they may:
  - Crack suddenly
  - Fracture rapidly
  - Shatter under high stress
- This is one of the main limitations of ceramics in biomedical implants.

Optical and Electrical Properties

- Ceramics can be:
  - **Opaque**
  - **Transparent**
- Some **dental ceramics** use these properties to **look similar to natural teeth.**
- Ceramics are also usually **good electrical** insulators.
- **Compared with metals**, ceramics have **very high electrical resistivity**.

Main Properties Summary (Fast review)

Property	Meaning
Hardness	Resist scratching and wear
High modulus	Stiff material
Compression strength	Strong under squeezing force
Brittleness	Can fracture suddenly
Chemical stability	Resist body fluid reactions
Electrical insulation	Poor electrical conductor
Optical behavior	Can be opaque or transparent

Classification of Ceramics

Classification of Ceramics

- **Ceramics** can be **classified** in **different** ways depending on their **structure** and use.
- The three main classification methods are:
  1. Based on form
  2. Based on composition
  3. Based on reactivity inside the body

Classification - Based on Form

- Based on their **physical form**, ceramics can be classified into:
  - **Powders**
  - **Coatings**
  - **Bulk shapes**
- Each form has a different biomedical function.



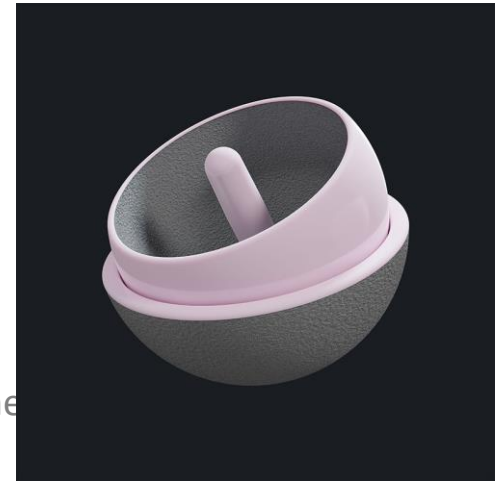


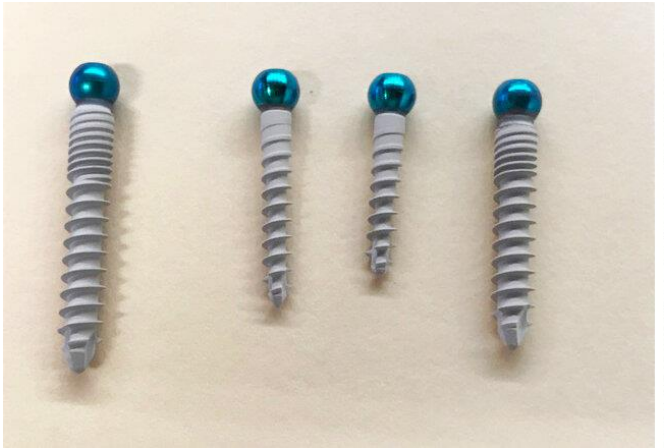
Ceramic Powders, Coatings, and Bulk Shapes

Form	Simple meaning	Biomedical example
Powders	Small dry solid particles	Bone fillers, calcium phosphate powders
Coatings	Thin ceramic layer on another material	Hydroxyapatite coating on titanium implants
Bulk shapes	Dense solid ceramic part	Dental crowns, ceramic hip components











Classification - Based on Composition

- Ceramics can also be classified according to their chemical composition:
  - **Oxide ceramics**
  - **Non-oxide ceramics**
  - **Composite ceramics**
- Composition affects:
  - Chemical stability
  - Electrical behavior
  - Thermal behavior
  - Mechanical performance

Oxide, Non-Oxide, and Composite Ceramics

Type	Main features	Examples
Oxide ceramics	Chemically stable, oxidation resistant, electrical insulators	Alumina, zirconia, magnesium oxide
Non-oxide ceramics	May have higher thermal/electrical conductivity, lower oxidation resistance	Carbides, nitrides, borides, silicates
Composite ceramics	Combination of different ceramic phases	Oxide + non-oxide ceramic systems

*Oxide ceramics such as **alumina and zirconia** are very important in dental and orthopedic applications.*

Oxide, Non-Oxide, and Composite Ceramics

Composition clue	Ceramic type	Examples
Contains O as the main element bonded with metal	**Oxide ceramic**	Al_2O_3, ZrO_2, MgO, TiO_2, SiO_2
Contains C, N, B and usually no oxygen	**Non-oxide ceramic**	SiC, B_4C, Si_3N_4, BN, TiB_2
Contains two or more ceramic phases/materials together	**Composite ceramic**	Al_2O_3–ZrO_2, HA–TCP, glass-ceramic composites

*Oxide ceramics such as **alumina and zirconia** are very important in dental and orthopedic applications.*

Classification - Based on Reactivity

- In biomaterials, the most important classification is based on how ceramics react inside the body.
- Ceramics can be classified as:
  - **Inert ceramics**
  - **Surface reactive ceramics**
  - **Resorbable ceramics**
- This classification is very important for biomedical engineering students.

Inert Ceramics

- Inert ceramics are chemically stable inside the body.
- They show little or no chemical change during long-term exposure to body fluids.
- Examples:
  - Alumina
  - Zirconia
  - Glass
  - Carbon-based ceramics
- Biomedical applications:
  - Hip joint components
  - Dental crowns
  - Dental implants



Surface Reactive and Resorbable Ceramics

Type	What happens inside the body?	Example	Biomedical meaning
Surface reactive ceramic	Reacts only at the surface	Bioactive glass	Forms a bone-like apatite layer
Resorbable ceramic	Gradually degrades/resorbs	Hydroxyapatite, tricalcium phosphate	Can be replaced by new bone





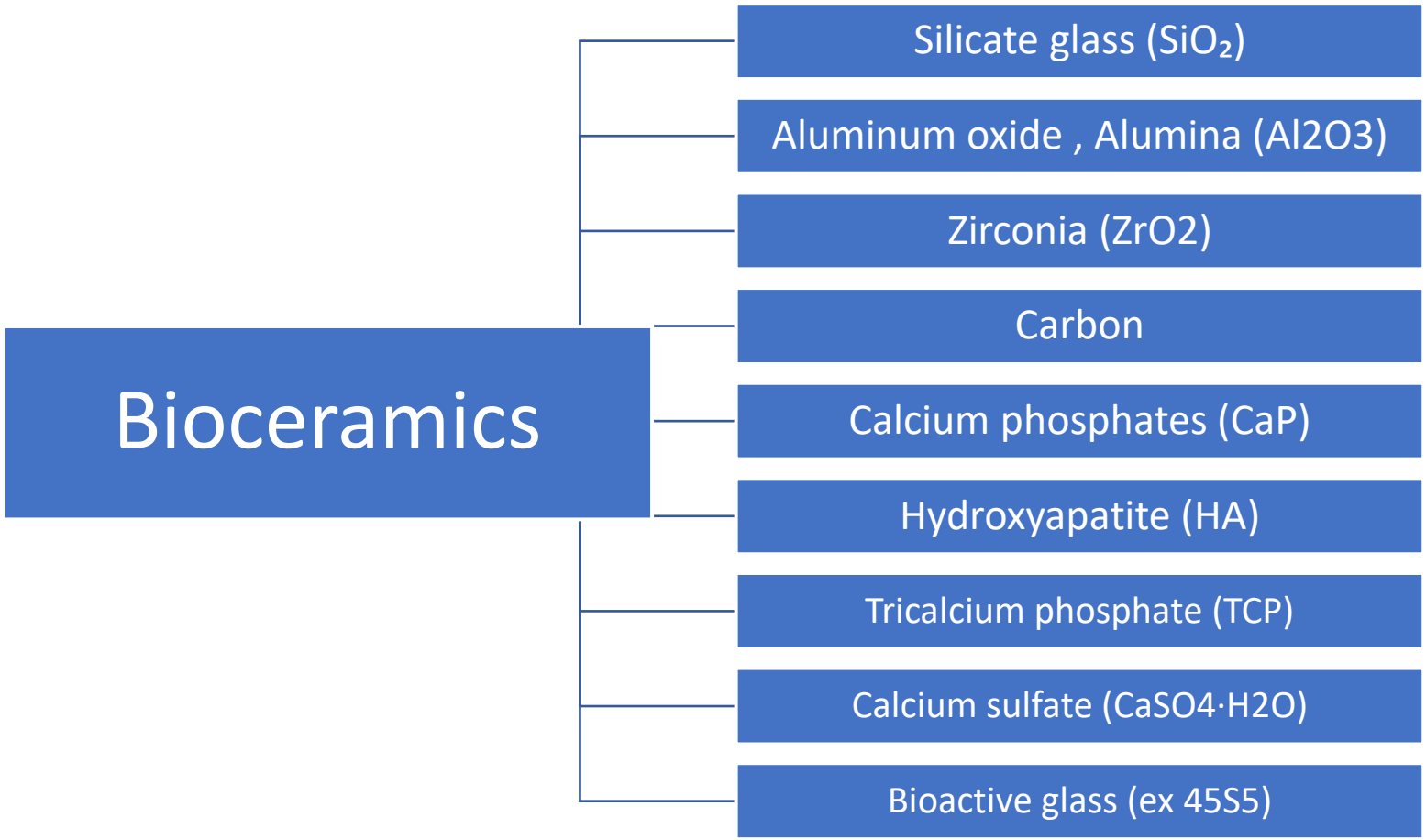
Summary of Ceramic Classifications

- Ceramics can be classified by:
  - **Form:** powders, coatings, bulk shapes
  - **Composition:** oxide, non-oxide, composite ceramics
  - **Reactivity:** inert, surface reactive, resorbable ceramics
- For biomedical applications, the most important question is:
  - **Does the ceramic stay stable, bond with tissue, or degrade over time?**
    - This determines its use in implants, coatings, dental materials, and bone repair.



Bioceramics

Bioceramics



Bioceramics

- **Bioceramics** are ceramic materials used in medical and dental applications.
- They can be used as:
  - Structural implants
  - Joint or tissue replacements
  - Coatings on metallic implants
  - Bone repair materials
  - Temporary scaffolds for tissue regeneration
  - Carriers for drugs, antibiotics, or growth factors



Requirements of Implantable Bioceramics

- Implantable bioceramics should be:
  - Non-toxic
  - Non-carcinogenic
  - Non-allergenic
  - Non-inflammatory
  - Mechanically stable for the intended use
  - Physically and chemically compatible with body fluids
  - Able to support tissue integration or regeneration when needed

Main Bioceramic Types in This Section

- The main bioceramics we need to know are:
  1. Silicate glass
  2. Alumina
  3. Zirconia
  4. Carbon / pyrolytic carbon
  5. Calcium phosphates
  6. Hydroxyapatite
  7. Tricalcium phosphate
  8. Calcium sulfate
  9. Bioactive glass

1) Silicate Glass (SiO_2)

- Silicate glass is mainly based on: **Silicon dioxide, SiO_2**
- SiO_2 is the basic glass-forming material.
- It forms a 3D network made of SiO_4 tetrahedra.
- Biomedical and medical-related uses include:
  - Ocular prostheses
  - Sensors
  - Optical fibers
  - Fiber lasers
  - Bioactive glass systems

Why Silicate Glass Matters

- Silicate glass is important because its properties can be changed by:
  - Chemical composition
  - Cooling rate
  - Network modifiers such as Na, Ca, Mg, Sr, or Ba oxides
- Changing the composition can affect:
  - Melting temperature
  - Viscosity
  - Chemical durability
  - Solubility
  - Optical properties

2) Alumina (Al_2O_3)

- **Alumina, Al_2O_3** Alumina is aluminum oxide.
- It is one of the most important inert bioceramics.
- Main properties:
  - Very hard
  - Scratch resistant
  - Chemically stable
  - Corrosion resistant in body fluids
  - Good wear resistance
  - Biocompatible



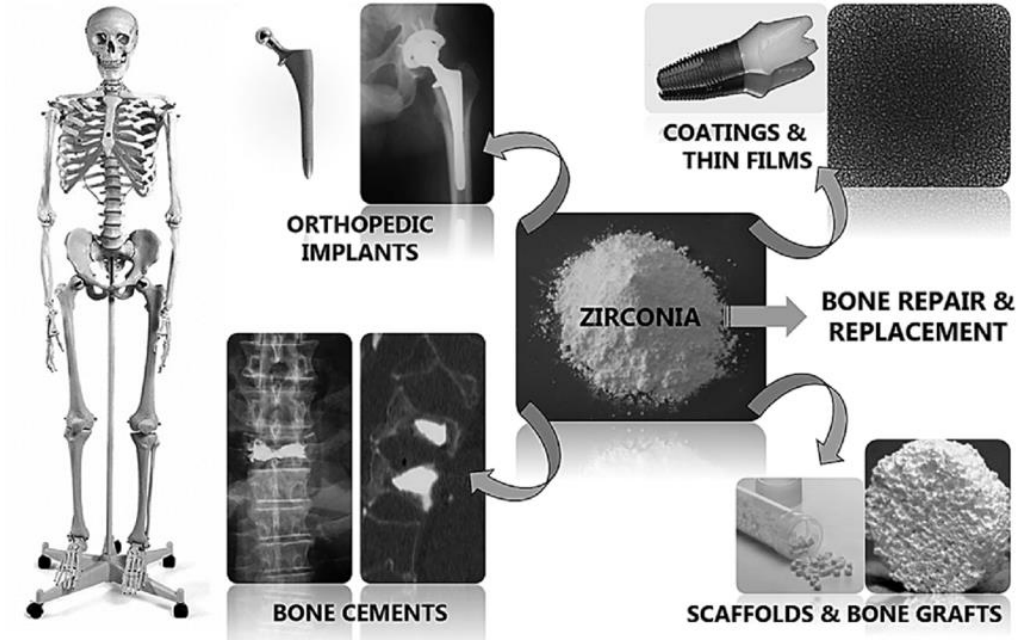
Biomedical Uses of Alumina

- Alumina is used mainly in:
  - Hip prostheses
  - Knee prostheses
  - Dental crowns
  - Dental ceramic restorations
- **Why useful?**
- Because it has excellent wear resistance and good biocompatibility.



3) Zirconia (ZrO_2)

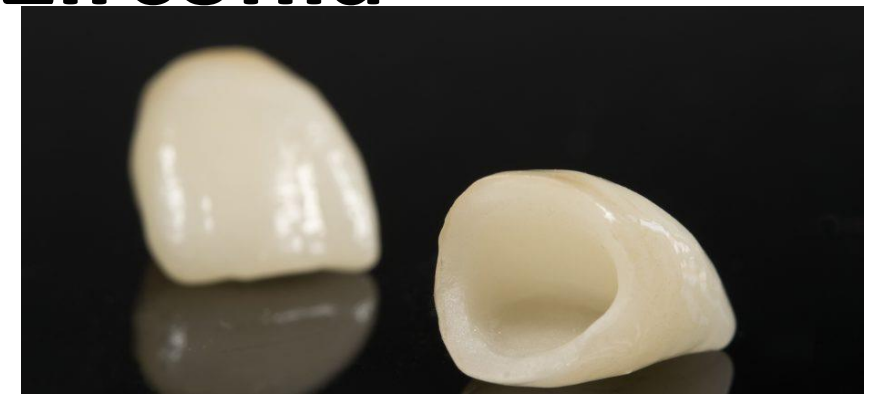
- Zirconia is zirconium oxide.
- It is a white crystalline ceramic used in medical and dental applications.
- Important properties:
  - High strength
  - High wear resistance
  - Good corrosion resistance
  - Better fracture toughness than alumina
  - Good esthetic appearance in dentistry



Why Zirconia Must Be Stabilized

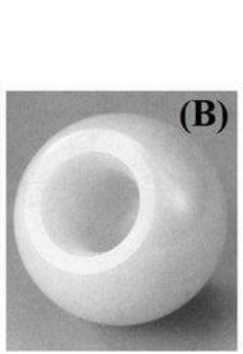
- Pure zirconia changes crystal structure with temperature.
- These phase changes can cause:
  - Volume expansion
  - Internal stresses
  - Cracking
- To avoid this, zirconia is stabilized using oxides such as:
  - Yttria, Y_2O_3
  - Magnesia, MgO

Biomedical Uses of Zirconia

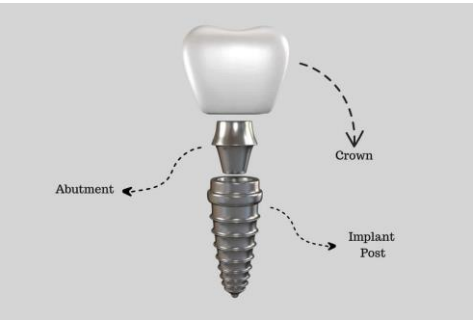


- Zirconia is used in:
  - Dental crowns
  - Dental bridges
  - Dental implant components
  - Hip implant ball heads
  - Ceramic subframes for dental restorations











4) Carbon as a Bioceramic

- Carbon can exist in different forms, such as:
  - Diamond
  - Graphite
  - Non-crystalline carbon
  - Pyrolytic carbon
- In biomaterials, the most important form is: **Pyrolytic carbon**

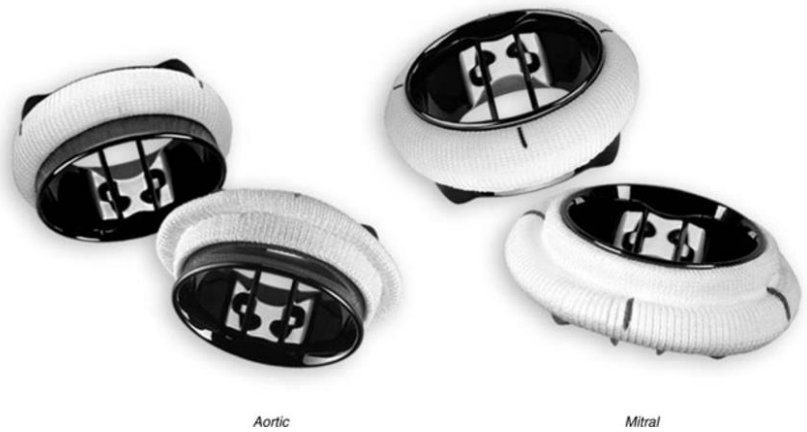


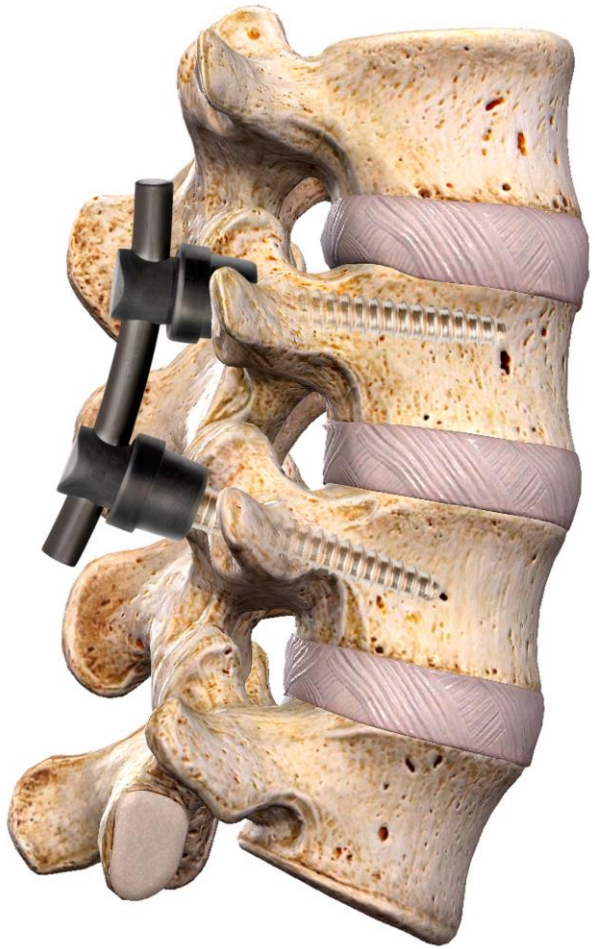
Pyrolytic Carbon

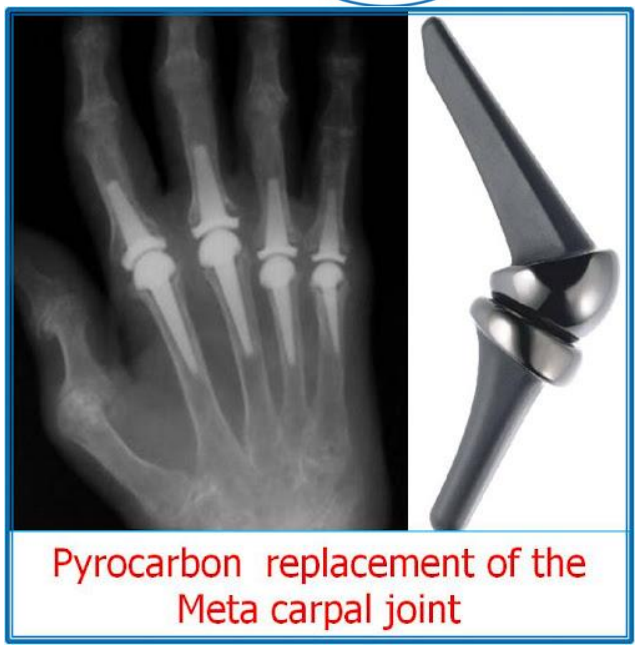
- Pyrolytic carbon is widely used in **implants** and **coatings**.
- Important properties:
  - Good biocompatibility
  - High durability
  - Good wear resistance
  - Good fatigue resistance
  - Thromboresistance
- **Thromboresistance means:** It has low tendency to cause blood clot formation.

Biomedical Uses of Pyrolytic Carbon

- Pyrolytic carbon is commonly used in:
  - Artificial heart valves
  - Heart valve coatings
  - Small joint implants
  - Finger joint implants
  - Spinal inserts

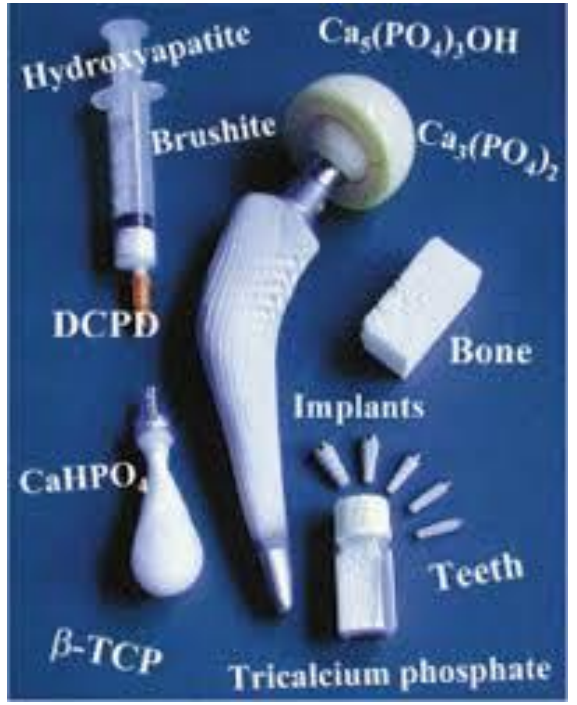





Pyrocarbon replacement of the Meta carpal joint

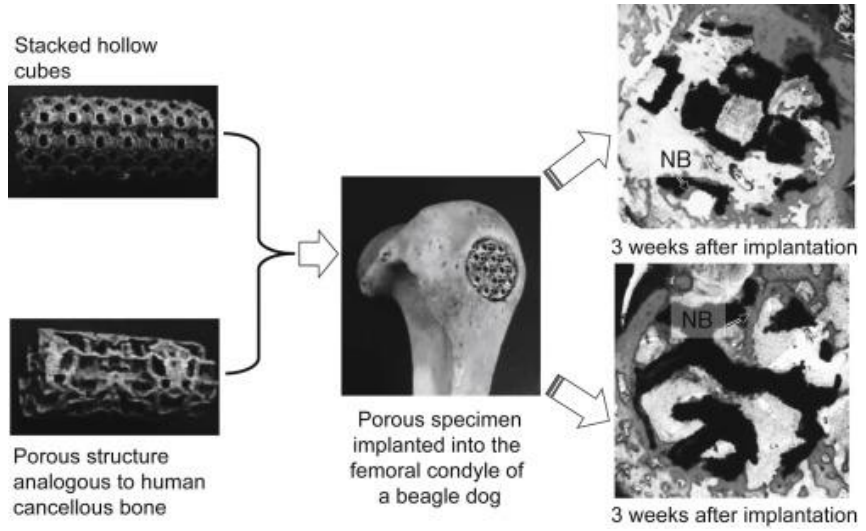
5) Calcium Phosphate Ceramics (CPCs)

- **Calcium phosphate ceramics** are very important in **bone-related biomaterials.**
- They are important because **bone mineral** naturally contains **calcium** and **phosphate**.
- Common types include:
  - Amorphous calcium phosphate, ACP
  - Brushite, DCPD
  - Octacalcium phosphate, OCP
  - **Tricalcium phosphate**, **TCP**
  - **Hydroxyapatite**, **HA**



Why Calcium Phosphates Are Important

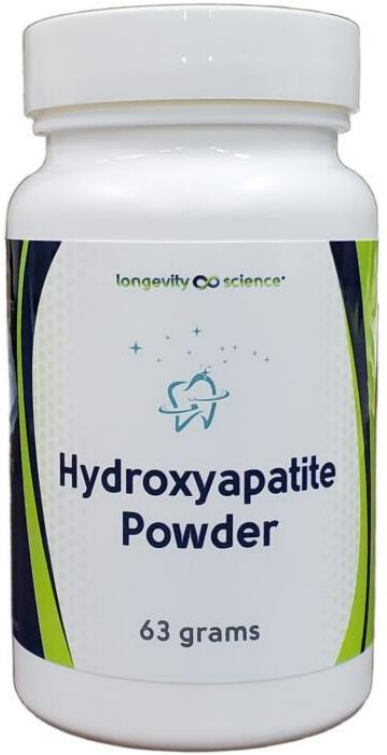
- Calcium phosphate ceramics are similar to biological apatites in bone and teeth.
- They can show:
  - Bioactivity
  - Biodegradation
  - **Osteoconductivity**
  - Bone bonding or bone support



- **Osteoconductivity means:** The material supports bone growth along its surface.

6) Hydroxyapatite

- **Hydroxyapatite, HA :** Hydroxyapatite is the most well-known calcium phosphate bioceramic.
- Chemical formula: $Ca_{10}(PO_4)_6(OH)_2$
- Important points:
  - **Similar** to **bone** and **tooth mineral**
  - Biocompatible
  - **Osteoconductive**
  - Supports **bone healing**
  - Helps improve **bone-implant bonding**

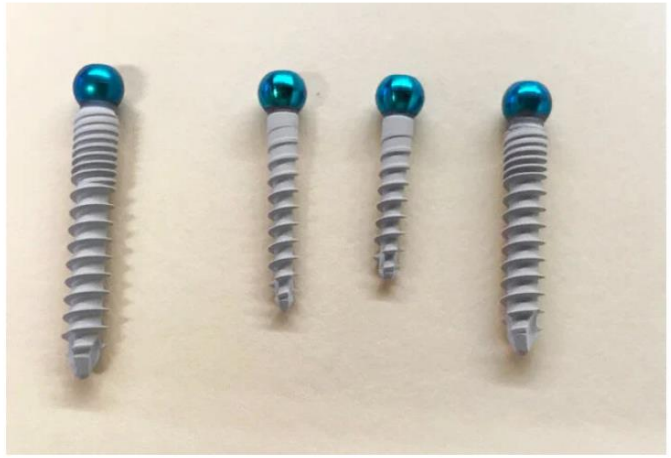


Biological Apatite vs Synthetic HA

- Pure synthetic **HA** is not **exactly** the **same** as **natural bone mineral**.
- Biological apatite may contain substituted ions such as:
  - Carbonate
  - Fluoride
  - Magnesium
  - Strontium

Why HA Is Used as a Coating

- HA is useful for improving bone attachment.
- However, HA is also:
  - Brittle
  - Strong in compression
  - Weak under tension and bending





Assorted titanium pedicle screws coated in hydroxyapatite via Himed's MATRIX HA™. These samples also include color anodized heads crafted by Himed to aid in simplified part identification.

- Because of this, HA is often used as a coating on metallic implants instead of being used as a large solid load-bearing implant.

7) Tricalcium Phosphate, TCP

- **Tricalcium phosphate** is another important **calcium phosphate** ceramic.
- Chemical formula:
  - **$Ca_3(PO_4)_2$**
- It exists in two main forms:
  - α-TCP
  - β-TCP
- **TCP** is **generally** more **soluble** than **HA**.



Biphasic Calcium Phosphate*

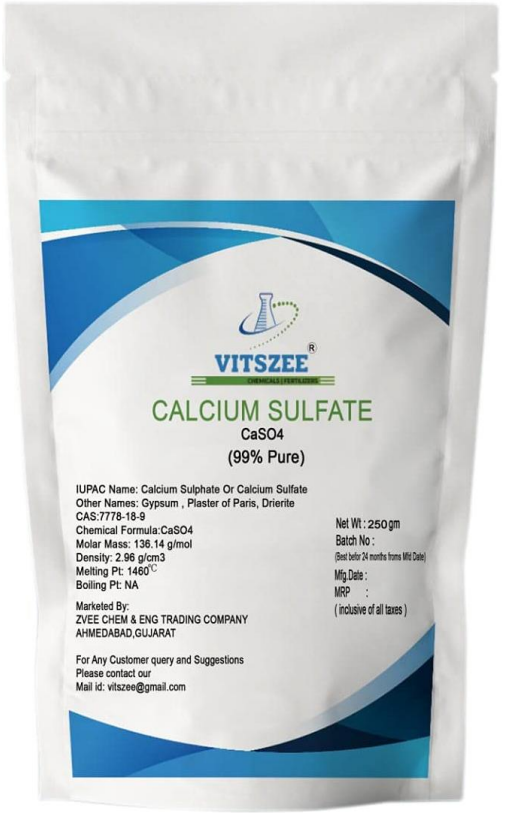
- **Biphasic calcium** phosphate contains:
  - **HA + TCP**
- It combines the advantages of both:
  - **HA** gives stability and **osteoconduction**
  - **TCP dissolves** and **releases calcium** and **phosphate** ions
- **More TCP** means **faster resorption**
- **More HA** means **slower resorption** and **longer support**

8) Calcium sulfate (CaSO4·H2O)

- **Calcium sulfate** is a resorbable **ceramic** used in **bone regeneration**.
- It is related to: **Plaster of Paris**
- Main advantages:
  - Biocompatible
  - Completely resorbable
  - Can be used as a bone defect filler



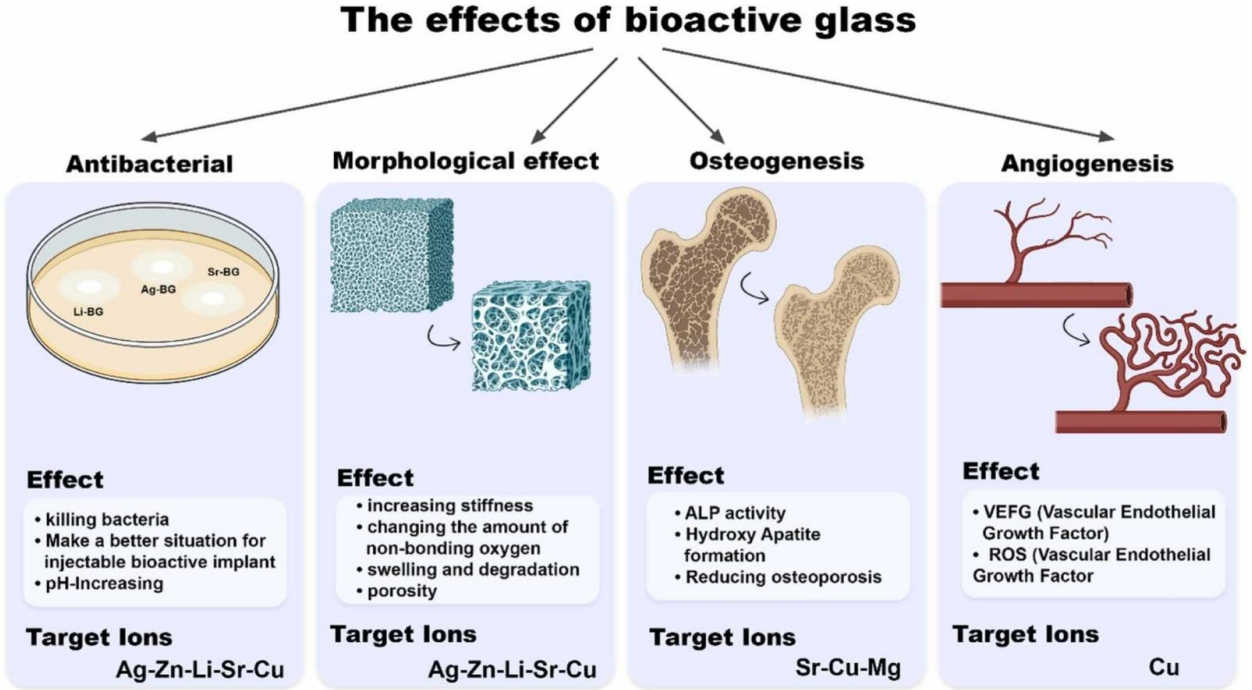




9) Bioactive Glass

- **Bioactive glass** is a glass designed to produce a biological response.
- It can bond to bone by forming a bone-like apatite layer on its surface.
- Common example: **45S5 bioactive glass**
- Main components:
  - SiO_2
  - P_2O_5
  - CaO
  - Na_2O
  - SrO





Biomedical Uses of Bioactive Glass



- **Bioactive glass** can be used as:
  - Bone defect filler
  - Granules or powders
  - Coatings on implants
  - Tissue engineering scaffolds
  - Dental and orthopedic repair materials
- Main advantage:
  - **It is surface reactive and can bond with bone.**

Bioceramic Medical Applications Summary

Bioceramic	Main feature	Main biomedical use
Silicate glass	Glass-forming SiO_2 network	Optical/medical glass, basis of bioactive glass
Alumina	Hard, inert, wear resistant	Hip/knee components, dental crowns
Zirconia	Strong, toughened ceramic	Dental restorations, hip ball heads
Pyrolytic carbon	Durable, thromboresistant	Artificial heart valves
HA	Bone-like, osteoconductive	Implant coatings, bone repair
TCP	More resorbable than HA	Bone fillers, temporary scaffolds
Calcium sulfate	Fast resorption	Bone defect filling
Bioactive glass	Forms bone-like apatite layer	Bone bonding, regeneration

Nanoceramics

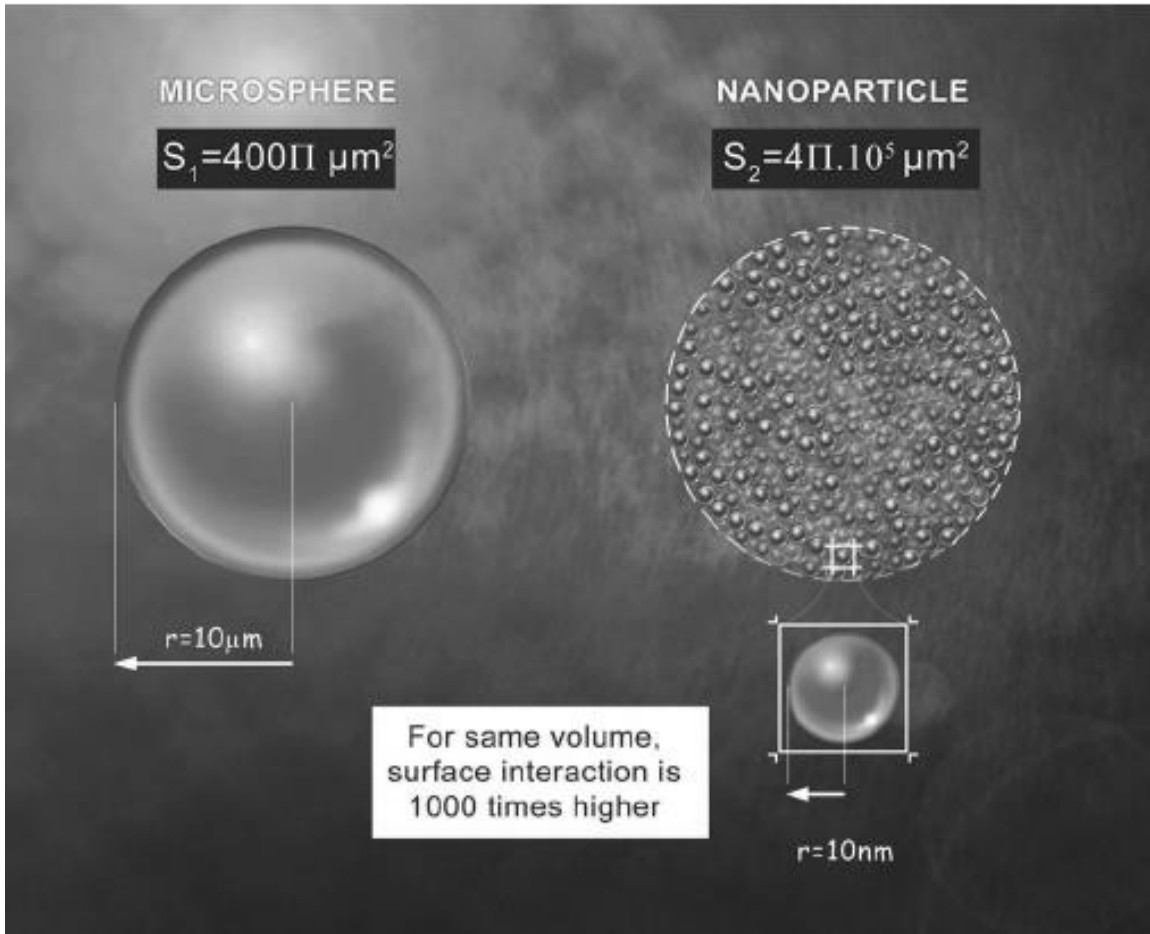
Nanoceramics

- Nanoceramics are ceramic materials with nanoscale size, usually **100 nm or less**.
- Biomedical applications include:
  - Imaging and diagnosis
  - Implantology
  - Drug delivery
  - Gene delivery
  - Cancer therapy
  - Tissue engineering



Why Nanosize Matters

- When ceramics become nanosized, they usually have:
  - Higher surface area
  - Higher surface area-to-volume ratio
  - More grain boundaries
  - Higher surface roughness
  - Higher hydrophilicity
  - Higher surface reactivity



Important Nanoceramic Examples

- Important nanoceramics include:
  - Nanophase alumina
  - Nano-zirconia
  - Alumina–zirconia nanocomposites
  - Carbon nanotubes
  - Nanophase calcium phosphate
  - Nano-hydroxyapatite
- These materials are studied to improve mechanical behavior, coatings, tissue regeneration, and drug delivery.

Nanoceramics in Bone and Tissue Engineering

- **Nanophase calcium phosphate** and **nano-hydroxyapatite** are important because **natural bone mineral** is also **nanoscale**.
- They can improve:
  - **Protein interaction**
  - **Osteoblast adhesion**
  - **Bone regeneration**
  - **Bone void filling**
  - **Injectable bone paste performance**

Summary of Ceramic Biomaterials

- **Ceramics** are **inorganic**, **non-metallic** materials mainly bonded by **ionic** and **covalent** bonds.
- **Strong bonds** make ceramics **hard**, **stiff**, **wear resistant**, **chemically stable**, and **heat resistant**.
- The **main limitation** of **ceramics** is **brittleness**, especially under **tension** or **bending**.
- **Ceramics** can be used **as bulk implants**, **coatings**, **powders**, **bone fillers**, **scaffolds**, and **dental materials**.

Summary of Ceramic Biomaterials

- **Bioceramics** can be **inert**, **surface reactive**, or **resorbable** depending on their behavior inside the body.
- Important examples include **alumina**, **zirconia**, **pyrolytic carbon**, **HA**, **TCP, calcium sulfate**, **bioactive glass**, and **nanoceramics**.
- In **biomedical engineering**, **ceramics** are especially important for **dental restorations**, joint components, bone repair, implant coatings, **heart valves**, and **tissue engineering**.

Think with me 🤔🤔🤔

- Why are ceramics usually hard and chemically stable, but also brittle?
- Compare inert, surface reactive, and resorbable ceramics with one biomedical example for each.
- Why is hydroxyapatite often used as a coating on metallic implants instead of as a large solid implant?
- Choose one bio-ceramic material from the lecture and explain its main biomedical use and why it is suitable for that application.



Thank you